4 Digital Technologies

Topic	What students need to learn:
4 Digital technologies	4.1 – Set-up, safe and accurate operation, advantages and disadvantages of the following digital technologies: a) computer-aided design (CAD) – 2D and 3D design to create and modify designs and create simulations, 3D modelling for creating 'virtual' products b) computer-aided manufacture (CAM) and rapid prototyping – CNC lathes, CNC routers, CNC milling machine, CNC laser, CNC vinyl cutters, rapid prototyping

Computer Aided Design

CAD

Computer Aided Design (CAD)…

CAD is the use of computers to develop 2D and 3D drawings and computer models with a range of file types that are often used to facilitate transfer to **CAM** equipment.

How CAD Changed Product Design – have a look at this website:

https://www.scan2cad.com/cad/cad-product-design/

Advantages and Disadvantages of CAD

Advantages	Disadvantages
The design and development process is faster and more accurate due to the advantages of editing and developing files rather than drawing by hand.	It is expensive to set up due to the hardware and software costs and training.
Networking facilities collaborative work across different offices and countries.	Software requires regular updating and ongoing technical support.
Highly accurate files can be sent to a range of CAM machines such as laser cutters, routers, laths and milling machines.	CAD is not always convenient for quick recording of ideas.
Files can be sent to 3D printers to make models or prototypes.	

Capabilities of 2D and 3D CAD Software…

- High-quality 3D CAD presentation drawings (‘artist impressions’) of designs can be created with photo-realistic rendering, thus creating ‘virtual’ products.
- Designs can be rotated and viewed from any angle.
- Client and marketing feedback is possible through renderings rather than physical models.
- ‘Virtual’ product ideas can be developed more quickly than with physical models and prototypes.
- Original artwork can be scanned into CAD software for further digital development.

Capabilities of 2D and 3D CAD Software…

- Editing tools (such as *'copy', 'mirror', 'rotate' and 'scale'*) accelerate the digital development of ideas.
- In industrial applications, CAD offers features such as:
  - *the ability to incorporate standard component files such as electronics and hydraulics in designs.*
  - *2D working drawings with multiple 'layers' to show various elements of a design, such as the structure, electronics and casing of a product, or plumbing and lighting circuitry in a building.*
  - *CAD/CAM simulation that can be executed to anticipate machining strategies and problems.*

Virtual Modelling…

- This is the use of 3D CAD files to simulate aspects of the performance of products before they are manufactured.
- Virtual models of mechanical designs such as car engines can be checked for fit and clearance.
- CAM processes can be simulated to ensure error-free machining.
- Production system planning is possible before investing in new equipment.
- A typical example of this would be the optimised layout of all elements of a car production system. This might include conveyors, workstations, robots, AGV's and particularly the flexible manufacturing system (FMS) cells required to carry out a wide range of operations.

Video: Jet Engine Cutaway 3D Animation…



Simulations…

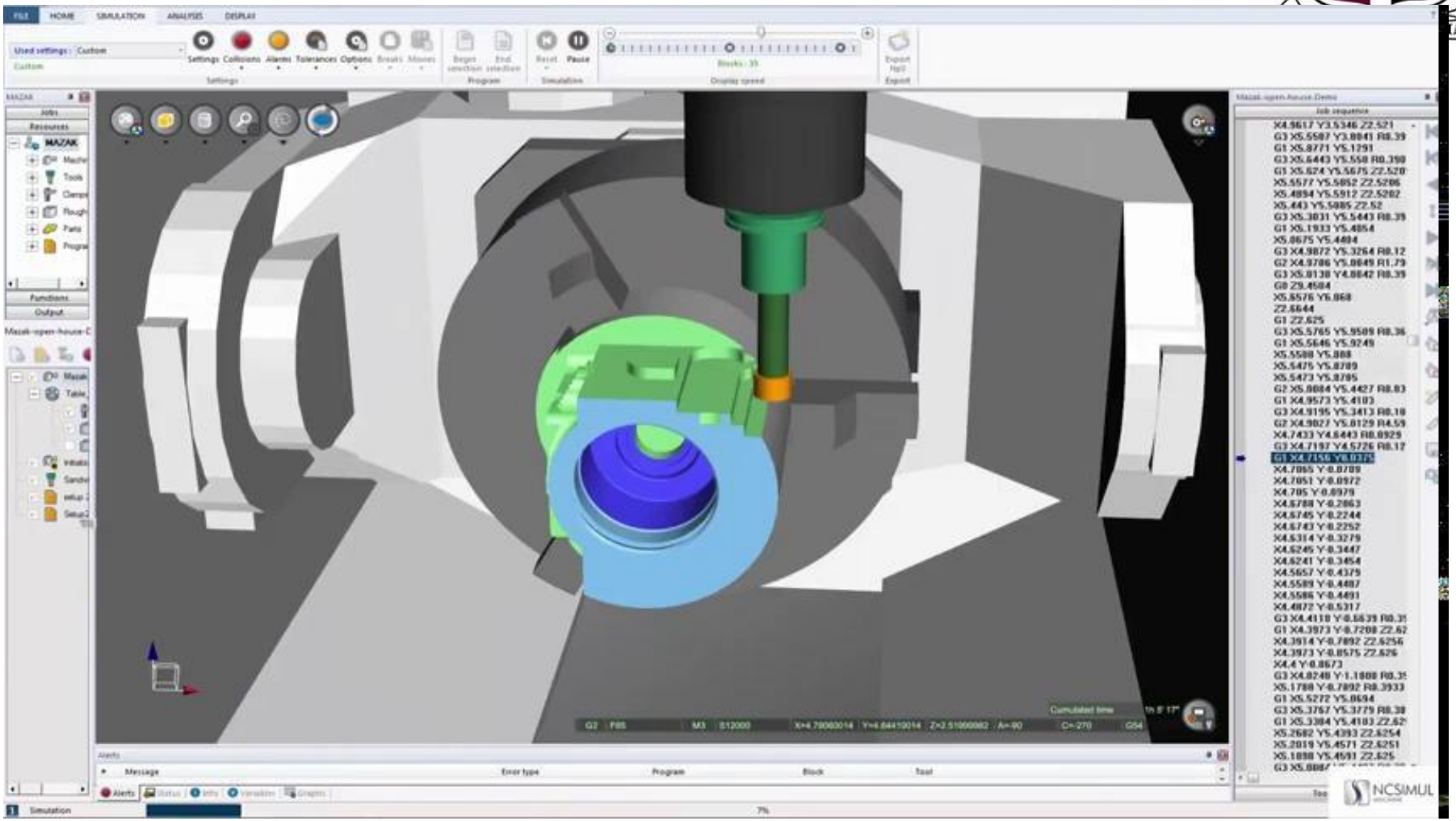
- A CAD/CAM simulation is a computer ‘run through’ of a CAM process which can be viewed by the designer, who also receives appropriate reports on its success.
- Examples include:
  - *The cutting path of a CNC laser*
  - *Various turning tools creating a profile for a part being made on a CNC lathe.*

Simulations…

Advantages:

- Highlights potential problems and shows if the production will be as expected.
- Indicates how long the item will take to be machined.
- Clashes with clamps and vices or the tool moving beyond the X, Y or Z limits will show up.
- Facilitates ‘nesting’ of multiple parts to reduce wastage of sheet material.
- 3D printer models’ appearance and support material requirements can be assessed.

Video: Realistic Machining Simulation…



Finite Element Analysis (FEA)…

- Finite Element Analysis (FEA) is an example of using CAD virtual modelling to test how a product component will withstand the conditions to which it is likely to be subjected.
- FEA analyses stress in elements of a CAD design.
- It allows designers to simulate forces, vibration or shock loads on products or components.
- Weak points are highlighted, usually using an easily identified colour scale.
- Money and time can be saved as designs can be optimised by modifying and re-testing.
- Static and dynamic testing is useful for many products, such as bicycle frames and cars.

Video: What is Finite Element Analysis…



Disadvantages of virtual modelling, simulations and FEA…

- Although these are very powerful and useful systems it is worth bearing in mind that they share some of the disadvantages mentioned when looking at CAD.
- They are expensive to set up due to hardware and software costs and training.
- Software requires regular updating and ongoing technical support.
- Software compatibility can sometimes be an issue that needs to be addressed when dealing with different companies' systems.

Exam Tip

If you are asked to explain the advantages of CAD compared to hand drawing, don't just state that you can use tools such as scale and copy.

Expand the point by mentioning that computer calculations ensure a level of accuracy and repeatability that is very difficult to achieve by manual methods.

Computer Aided Manufacture

CAM

Video: The History of CNC…



Computer Aided Manufacture (CAM)…

CAM Processes:

- CAD files are downloaded to computer numerically controlled (CNC) machines for cutting and shaping materials.
- The machine generates a control program, facilitating settings such as tool selection, feed rate, cutter speed and other factors to suit the material being machined.
- The simple and seamless transfer of files from CAD to CAM also makes one-off production viable.

Computer Aided Manufacture (CAM)…

Computer Numerical Control (CNC):

- This is the use of a program, converted from CAD files, which uses special codes to control CAM equipment.
- Examples of CNC machines include:
  - *CNC Laser Cutting*
  - *CNC Routing*
  - *CNC Milling*
  - *CNC Turning*
  - *CNC Plotter Cutting*

Computer Aided Manufacture (CAM)…

CNC Laser Cutting

- 2D CAD files are converted into a CNC program.
- The X and Y axes, speed and power, are programmed to cut and engrave a range of materials.
- Lasers use a high-energy laser beam to cut or vaporise materials
- Compressed gas or air blows the waste material away to leave a clean edge.

Computer Aided Manufacture (CAM)…

CNC Routing

- CNC routers use high-speed cutters to machine 2D CAD files in sheet materials such as MDF.
- 3D CAD files can also be used to machine models, moulds and other 3D forms in materials such as timber and modelling foams.
- The cutter moves in the X, Y and Z axes, and the work piece is fixed to the bed of the machine.

Computer Aided Manufacture (CAM)…

CNC Milling

- Milling machines move the work piece in the X, Y and Z axes, and sometimes have two or three additional axes of movement for the cutting head and work piece.
- Milling machines are more robust than routers with slow speeds to suit metal cutting.
- Tools can be automatically selected to drill holes, cut slots and shape complex surfaces and edges.

Computer Aided Manufacture (CAM)…

CNC Milling

- These machines are usually totally enclosed to improve safety.
- Coolant improves tool life and quality of finish by providing lubrication and reducing heat.
- CNC milling machines, as with the other types of CNC machine, operate much more quickly, accurately and with greater endurance than human operators, needing to stop only for essential maintenance.

Computer Aided Manufacture (CAM)…

CNC Turning

- CNC lathes usually turn (machine) metals and plastics in bar form to create pre-programmed X and Y axis profiles, including threads and bored recesses.
- They can automatically change tools, which often use disposable tungsten carbide tips.
- CNC lathes are often teamed with milling machines and robot arms in FMS cells.

Computer Aided Manufacture (CAM)…

CNC Plotter Cutting

- Plotter cutters are used to cut out shapes in compliant materials such a self adhesive vinyl.
- They generate an X and Y co-ordinate cutting path from 2D CAD files.
- The pieces produced are often used for graphics applications.
- Plotter cutters usually pull the sheet material in and out of the machine (Y axis), while the cutter moves from side to side in the X axis.



Rapid Prototyping Processes

Rapid Prototyping Processes…

- Rapid prototyping is the use of machines to produce realistic 3D models from CAD files.
- The models can be made in a range of materials, which are usually a type of polymer.
- The benefits to designers and manufacturers include:
  - *It is a fast process, as 3D modelling needs minimal human input*
  - *Modelling was previously highly skilled, expensive and time consuming*
  - *Rapid prototypes can serve as models for casting and making production dies*

Rapid Prototyping Processes…

- *Stereolithography modelling, where a UV laser solidifies layers of photopolymer (light sensitive resin), is often used by manufacturers such as potteries*
- *Stereolithography models o f pottery designs can be realistically sprayed and decorated for appraisal by designers and potential customers*

Fused Deposition Modelling (FDM)/3D Printing…

- 3D printers use FDM to build, or ‘print’, a 3D model from 3D CAD files.
- The 3D CAD file model is ‘sliced’ into layers that are extruded to build the product from the bottom up, layer by layer.
- FDM uses a very thin extruded thermoplastic filament, and sometimes an additional support material, to build the model.
- The supporting ‘scaffold’ is broken away or chemically dissolved after manufacture.

Fused Deposition Modelling (FDM)/3D Printing…

- 3D printers are now being used for one-off production of bespoke items such as jewellery cast in a mould created using a 3D printed pattern.
- Customised grips for people with disabilities can also be made from 3D printed ABS.
- 3D printing of metals can produce highly complex bespoke engineering components that would be impossible to make in any other way and have no potentially weak joints.

Fused Deposition Modelling (FDM)/3D Printing…

- Titanium, stainless steel and aluminium parts can be made with metal 3D printers.
- 3D printers for metal use an electron or laser beam to fuse metal powder layer by layer.
- Custom-made prosthetics, joints, bone replacements and other medical products, can be 3D printed on titanium.

Video: The Lily Light…



Typical Mistake

Don’t confuse the terms ***‘rapid prototype’*** and ***‘quick prototype’***.

The term ***‘rapid prototype’*** has become synonymous with models made using ***computer-based-technology*** such as ***3D printers.***

‘Quick prototype’ is more likely to refer to the use of *modelling foam, card, wire and other similar materials* to mock-up a potential design.

Now Test Yourself

1. *Give **four** advantages of CAD over traditional drawing techniques.*
2. *What is a **rapid prototype**?*
3. *Why might a 2D CAD drawing of a building be useful for an architect to communicate appropriate information to various trades people who may work on a project?*
4. ***Name** and **briefly describe** a method of virtual testing to anticipate how product parts will perform?*
5. *What is **CNC milling**?*